

5.1.1

$$a) \quad \frac{\dot{x}}{\dot{v}} = \frac{v}{-\omega^2 x} \Rightarrow -\omega^2 x \dot{x} = v \dot{v}$$

$$\int -\omega^2 x \frac{dx}{dt} = \int v \frac{dv}{dt}$$

$$\text{This implies } \omega^2 x^2 + v^2 = C$$

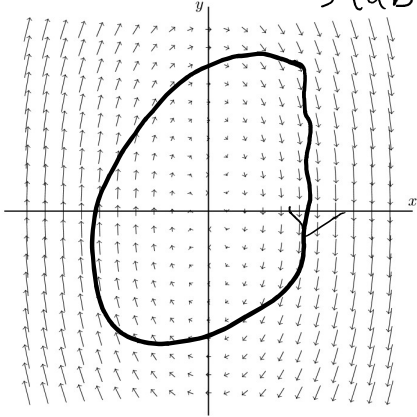
b) if $C = 2E/m$ Then we get

$$\frac{1}{2} m \omega^2 x^2 + \frac{1}{2} m v^2 = E$$

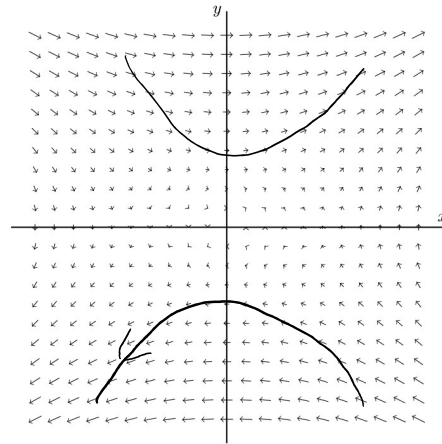
This is just the total energy
for a simple harmonic oscillator

5.1.10

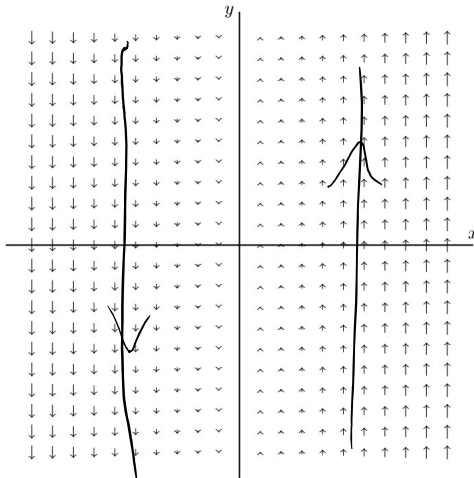
a) Lyapunov stable



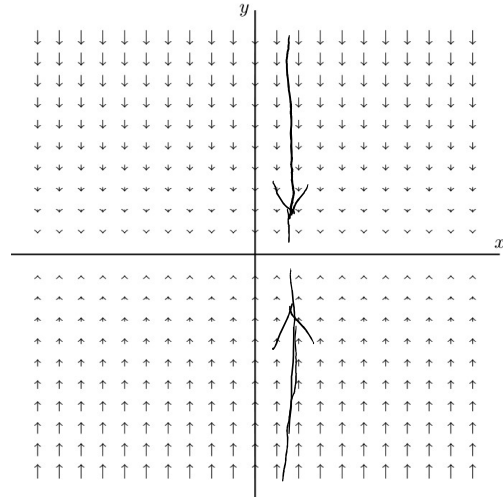
b) none of the above



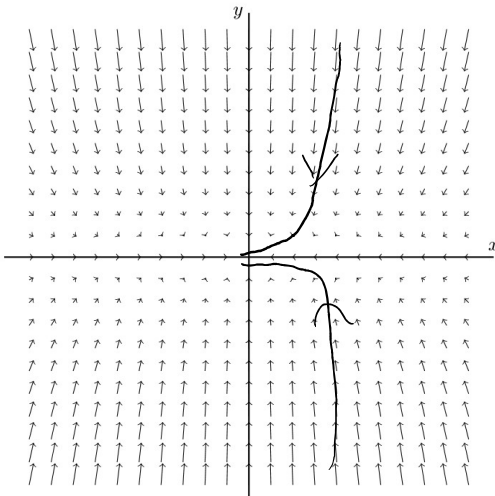
c) none of the above



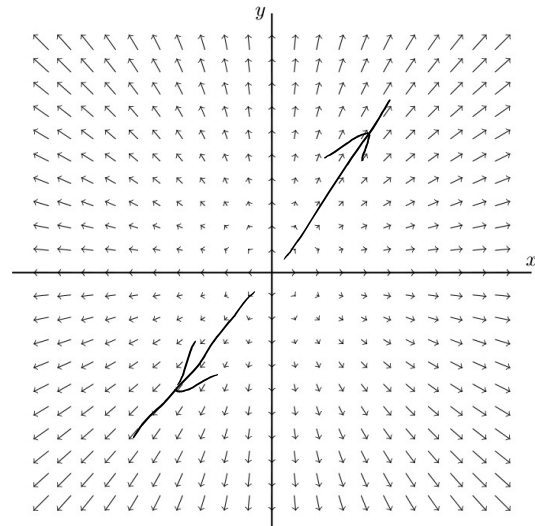
d) Lyapunov stable



e) Asymptotically stable



f) none of the above



5.2.12 a.) For some reason, Strogatz uses unconventional notation in this problem, using the symbol I for charge instead of Q . Starting from

$$L\ddot{Q} + R\dot{Q} + Q/C = 0$$

Let $I = \dot{Q}$ then the 2nd order ODE can be rewritten as the 2D coupled system

$$\left. \begin{aligned} \dot{Q} &= I \\ \dot{I} &= -\frac{R}{L}I - \frac{1}{LC}Q \end{aligned} \right\} \text{ has fixed point at } (0,0)$$

b.)

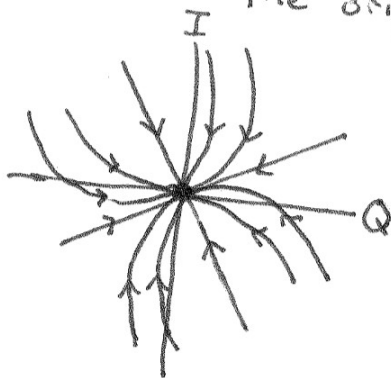
$$A = \begin{pmatrix} 0 & 1 \\ -\frac{1}{LC} & -\frac{R}{L} \end{pmatrix} \bigg|_{0,0} \Rightarrow \begin{aligned} \tau &= -\frac{R}{L} \\ \Delta &= \frac{1}{LC} \end{aligned}$$

Since $\tau < 0$ the fixed point $(0,0)$ is asymptotically stable for $R > 0$. If $R = 0$ then $\tau = 0$ and $\Delta > 0$ says that $(0,0)$ is a center. Centers are Lyapunov stable but not attracting so they are neutrally stable. Therefore, the origin is neutrally stable for $R = 0$.

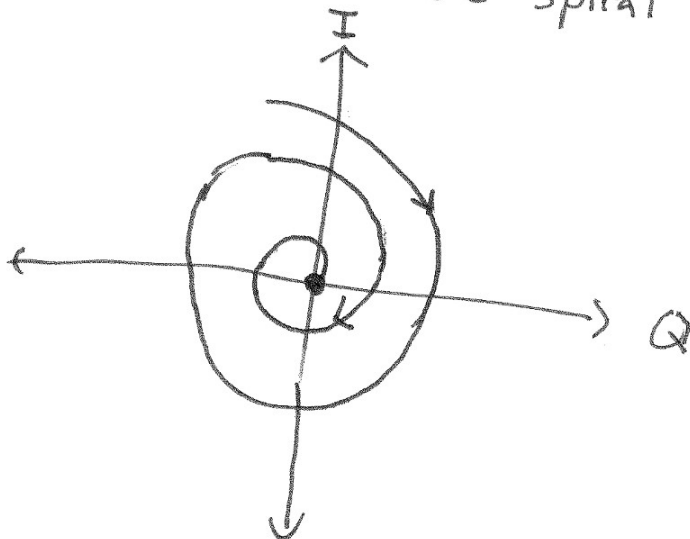
S.2.12
cont.

$$\begin{aligned} c.) \quad \lambda_{1,2} &= \frac{-\frac{R}{L} \pm \sqrt{\frac{R^2}{L^2} - 4\left(\frac{1}{LC}\right)}}{2} = -\frac{R}{2L} \left(1 \pm \sqrt{1 - \frac{4L}{R^2C}} \right) \\ &= \frac{-\frac{R}{L} \pm \sqrt{\frac{1}{L^2C}(R^2C - 4L)}}{2} \end{aligned}$$

If $R^2C > 4L$, then both eigenvalues are real and negative so the origin is a stable node.



If $R^2C < 4L$, the term in the radical is negative and λ_1 and λ_2 are a complex conjugate pair with a negative real part, so the origin is a stable spiral.



* For $Q_0, I_0 > 0$
 $\dot{Q} > 0$ and $\dot{I} < 0$
so spiral rotates clockwise

5.2.12
cont.

If $R^2C = 4L$ then $\lambda_{1,2} = -\frac{R}{2L} = -\frac{2}{RC}$ $\Leftarrow$ stable degenerate case! $\lambda \neq 0$ (star, d. node)

$$A = \begin{pmatrix} 0 & 1 \\ -\frac{1}{LC} & -\frac{R}{L} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{4}{R^2C^2} & -\frac{4}{RC} \end{pmatrix} \quad \text{so that}$$

the eigenvectors are given by

$$\begin{pmatrix} 0 & 1 \\ -\frac{4}{R^2C^2} & -\frac{4}{RC} \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = -\frac{2}{RC} \begin{pmatrix} v_x \\ v_y \end{pmatrix}$$

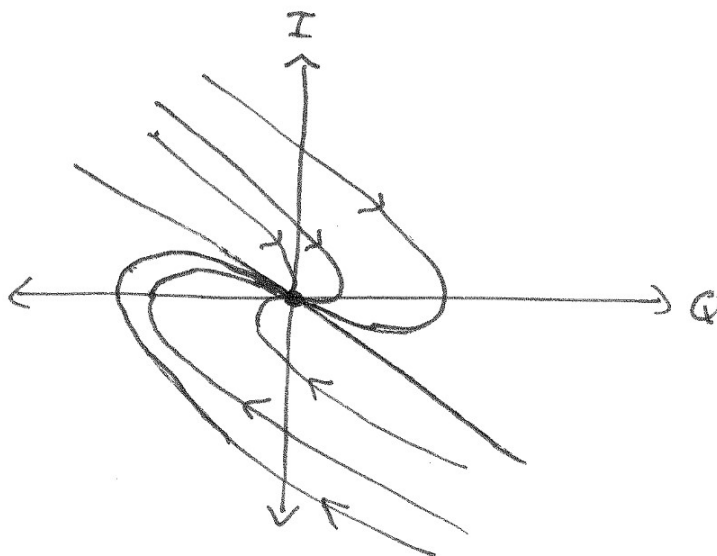
$$\Rightarrow v_y = -\frac{2}{RC} v_x$$

$$-\frac{4}{R^2C^2} v_x - \frac{4}{RC} v_y = -\frac{2}{RC} v_y$$

$$= -\frac{4}{R^2C^2} v_x + \frac{8}{R^2C^2} v_x = -\frac{4}{R^2C^2} v_x$$

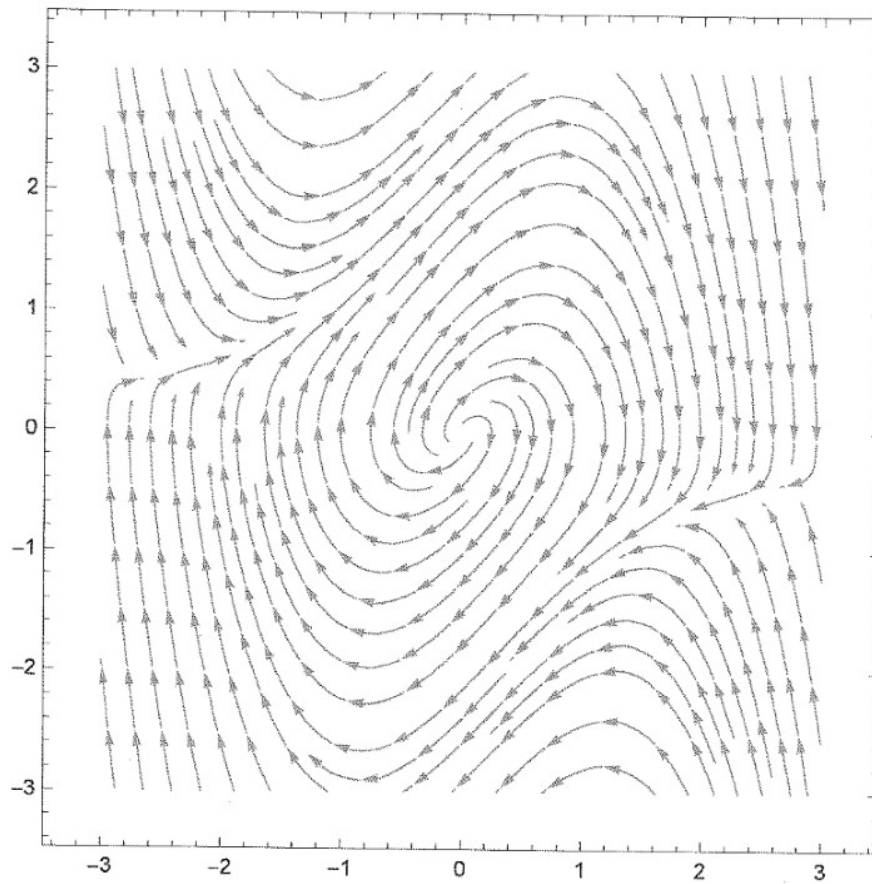
Therefore the only eigenvector is $\begin{pmatrix} RC/2 \\ -1 \end{pmatrix}$

So the origin is a stable degenerate node.



6.1.8

```
StreamPlot[{y, -x + y*(1 - x^2)}, {x, -3, 3}, {y, -3, 3}]
```



6.2.1

$$\begin{aligned}\dot{x}_1 &= f_1(x_1, x_2, x_3, \dots, x_n) \\ \dot{x}_2 &= f_2(x_1, \dots, x_n) \\ &\vdots \\ \dot{x}_i &= f_i(x_j)\end{aligned}$$

If The dynamical system satisfies the Existence and Uniqueness Theorem in Section 6.2, Then $f_i(x_j)$ can be expressed as a polynomial (e.g. can be Taylor expanded). (An example of a system that does not satisfy this is given in example 2.5.1.) For a system that does satisfy the theorem, $f_j(x_i + \epsilon_i)$ can be expressed as

$$f_j(x_i + \epsilon_i) = g_{0,j}(x_i)\epsilon^0 + g_{1,j}(x_i)\epsilon^1 + g_{2,j}(x_i)\epsilon^2 + \dots + g_{n,j}(x_i)\epsilon^n$$

Where ϵ_i is a small perturbation. Now, Let x^* be a fixed point ($f_i(x_i^*) = 0$).

If $x_i = x_i^*$, Then

$$\begin{aligned}\frac{d}{dt}(x_i^* + \epsilon_i) &= g_{0,j}(x_i^*) + \sum_{n=1}^{\infty} g_{n,j}(x_i^*)\epsilon^n \\ \dot{x}_i^* + \dot{\epsilon}_i &= g_{0,j}(x_i^*) + \sum_{n=1}^{\infty} g_{n,j}(x_i^*)\epsilon^n \\ \dot{\epsilon}_i &= \sum_{n=1}^{\infty} g_{n,j}(x_i^*)\epsilon^n\end{aligned}$$

If ϵ_i is small, ϵ^n will be even smaller. This means we can approximate the dynamics by truncating this sum at $n=1$.

$$\dot{\epsilon}_i \approx g_{1,j}(x_i^*)\epsilon$$

The solutions to this equation are

$$\epsilon_i(t) = A_i e^{g_{1,j}(x_i^*)t}$$

for a fixed point attractor, $g_{1,j}(x_i^*) < 0$, so $\epsilon \rightarrow 0$ as $t \rightarrow \infty$.

This means that, for this type of system, the trajectories never actually reach the point x_i^* and therefore never actually intersect.

6.3.11 a) $\dot{r} = -r \Rightarrow r(t) = r_0 e^{-t}$

$$\dot{\theta} = \frac{1}{\ln[r]} = \frac{1}{\ln[r_0 e^{-t}]} = \frac{1}{\ln[r_0] + \ln[e^{-t}]}$$

$$\dot{\theta} = \frac{1}{\ln[r_0] - t} \Rightarrow \int d\theta = \int \frac{1}{\ln[r_0] - t} dt$$

$$\theta(t) = \ln \left[\frac{1}{|\ln[r_0] - t|} \right] + \theta_0$$

b) $\lim_{t \rightarrow \infty} r(t) = \lim_{t \rightarrow \infty} r_0 e^{-t} = 0$ so, stable

$$\lim_{t \rightarrow \infty} \theta(t) = \lim_{t \rightarrow \infty} \ln \left[\frac{1}{|\ln[r_0] - t|} \right] + \theta_0 = \infty$$

so, θ grows and grows and grows

c) $\dot{r} = \frac{d}{dt} [\sqrt{x^2 + y^2}] = \frac{x\dot{x} + y\dot{y}}{\sqrt{x^2 + y^2}} = -r = -\sqrt{x^2 + y^2}$

$$\dot{\theta} = \frac{d}{dt} \left[\tan^{-1} \left[\frac{y}{x} \right] \right] = \frac{x\dot{y} - y\dot{x}}{x^2 + y^2} = \frac{1}{\ln[r]} = \frac{2}{\ln[x^2 + y^2]}$$

① $\frac{x\dot{x} + y\dot{y}}{\sqrt{x^2 + y^2}} = -\sqrt{x^2 + y^2}$ and ② $\frac{x\dot{y} - y\dot{x}}{x^2 + y^2} = \frac{2}{\ln[x^2 + y^2]}$

$$x\dot{x} + y\dot{y} = -(x^2 + y^2)$$

$$x\dot{y} - y\dot{x} = \frac{2(x^2 + y^2)}{\ln[x^2 + y^2]}$$

① $x(x\dot{x} + y\dot{y}) - y(x\dot{y} - y\dot{x}) = \dot{x}(x^2 + y^2) = -x(x^2 + y^2) - y \frac{2(x^2 + y^2)}{\ln[x^2 + y^2]} \Rightarrow \dot{x} = -x - \frac{2y}{\ln[x^2 + y^2]}$

② $y(x\dot{x} + y\dot{y}) + x(x\dot{y} - y\dot{x}) = \dot{y}(x^2 + y^2) = y(x^2 + y^2) + x \frac{2(x^2 + y^2)}{\ln[x^2 + y^2]} \Rightarrow \dot{y} = -y + \frac{2x}{\ln[x^2 + y^2]}$

6.3.11 (cont.)

$$d) \quad J = \begin{pmatrix} \frac{4xy}{(x^2+y^2)\ln^2[x^2+y^2]} - 1 & \frac{4y^2}{(x^2+y^2)\ln^2[a^2+y^2]} - \frac{2}{\ln[a^2+y^2]} \\ \frac{2}{\ln[x^2+y^2]} - \frac{4x^2}{(x^2+y^2)\ln^2[x^2+y^2]} & \frac{-4xy}{(x^2+y^2)\ln^2[a^2+y^2]} - 1 \end{pmatrix}$$

$$J_{(0,0)} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \Rightarrow \begin{aligned} \dot{x} &= -x \\ \dot{y} &= -y \end{aligned} \Rightarrow \text{stable}$$

6.4.7 In order to simplify the analysis, we first nondimensionalize the problem so that we don't have to deal with seven parameters.

Let $n_1 = Xx$, $n_2 = Yy$, and $t = T\tau$, so that

$$\frac{Xx'}{T} = G_1 Xx (N_0 - \alpha_1 Xx - \alpha_2 Yy) - K_1 Xx$$

$$\frac{Yy'}{T} = G_2 Yy (N_0 - \alpha_1 Xx - \alpha_2 Yy) - K_2 Yy, \text{ so}$$

$$x' = G_1 T N_0 x \left(1 - \frac{\alpha_1}{N_0} Xx - \frac{\alpha_2}{N_0} Yy\right) - K_1 T x$$

$$y' = G_2 T N_0 y \left(1 - \frac{\alpha_1}{N_0} Xx - \frac{\alpha_2}{N_0} Yy\right) - K_2 T y$$

Let $X = \frac{N_0}{\alpha_1}$, $Y = \frac{N_0}{\alpha_2}$ and $T = \frac{1}{G_1 N_0}$, then

$$x' = x(1 - x - y) - \frac{K_1}{G_1 N_0} x = \left(1 - \frac{K_1}{G_1 N_0}\right)x - (x^2 + xy)$$

$$y' = \frac{G_2}{G_1} y(1 - x - y) - \frac{K_2}{G_1 N_0} y = \left(\frac{G_2 N_0 - K_2}{G_1 N_0}\right)y - \frac{G_2}{G_1}(y^2 + xy)$$

Let $A = 1 - \frac{K_1}{G_1 N_0}$, $B = \frac{G_2 N_0 - K_2}{G_1 N_0}$ and $C = \frac{G_2}{G_1}$

$$x' = Ax - (x^2 + xy)$$

$$y' = By - C(y^2 + xy)$$

The Jacobian for this system is

$$A = \begin{pmatrix} A - 2x - y & -x \\ -Cy & B - 2Cy - Cx \end{pmatrix}$$

a.) At $(0,0)$

$$A = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \Rightarrow \lambda_1 = A = 1 - \frac{k_1}{G_1 N_0} = \frac{G_1 N_0 - k_1}{G_1 N_0} \rightarrow v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$\lambda_2 = B = \frac{G_2 N_0 - k_2}{G_1 N_0} \rightarrow v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Therefore the origin is a stable node when $A, B < 0$ and the laser just acts as a lamp.

Note, that for this case it is the only physical attractor since the other fixed points $(A, 0)$ and $(0, B/c)$ (see part b) would imply negative photon populations.

For $A > 0, B < 0$ the origin will be a saddle whose stable manifold is the y axis and whose unstable manifold is the x -axis. Similarly, when $A < 0, B > 0$ the origin will be a saddle but the stability of the axes will be reversed.

Finally, when both A and $B > 0$, the origin will be an unstable node.

Additional, borderline configurations exist but these are not structurally stable and would never be observed in a real laser system.

b.) For $x=0$, $x'=0$ and $y' = By - Cy^2 \rightarrow (x^*, y^*) = (0, B/c)$

For $y=0$, $y'=0$ and $x' = Ax - x^2 \rightarrow (x^*, y^*) = (A, 0)$

At $(x^*, y^*) = (A, 0)$

$$A = \begin{pmatrix} -A & -A \\ 0 & B-CA \end{pmatrix} \rightarrow \left. \begin{array}{l} \tau = B - A(1+C) \\ \Delta = CA^2 - BA \end{array} \right\} \begin{array}{l} \lambda_1 = B - AC \\ \lambda_2 = -A \end{array}$$

At $(x^*, y^*) = (0, B/c)$

$$A = \begin{pmatrix} A - B/c & 0 \\ -B & -B \end{pmatrix} \rightarrow \left. \begin{array}{l} \tau = A - \frac{B}{c}(c+1) \\ \Delta = \frac{B^2}{c} - BA \end{array} \right\} \begin{array}{l} \lambda_1 = -B \\ \lambda_2 = A - B/c = -\frac{1}{c}(B-CA) \end{array}$$

Note that since x and y are dimensionless populations these fixed points only exist when $A, B \geq 0$ (C is always positive since it is a ratio of the cavity gains).

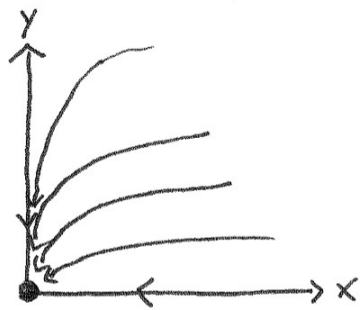
For $B > AC$, $(A, 0)$ is a saddle and $(0, B/c)$ is a stable node.

For $B < AC$, $(0, B/c)$ is a saddle and $(A, 0)$ is a stable node.

The case where $B = AC$ is structurally unstable and therefore will never be seen in a real laser.

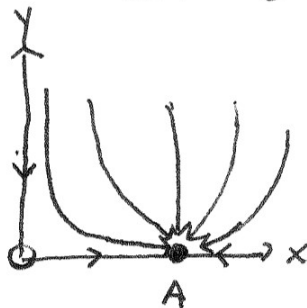
c.) For the physically relevant cases:

i.) $A < 0$ ($G_1 < k_1/N_0$) and $B < 0$ ($G_2 < k_2/N_0$)



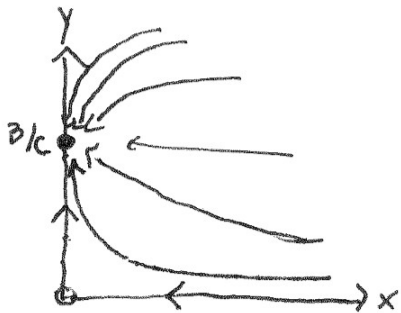
The only physically relevant fixed point is the origin and it is a stable node

ii.) $A > 0$ and $B < 0$ ($G_1 > k_1/N_0$ & $G_2 < k_2/N_0$)



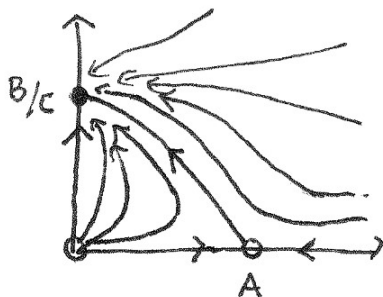
The only attractor is the fixed point $(A, 0)$ and after an initial transient involving both modes the laser settles to operation in the first (n_1) mode. The number of n_2 photon decreases monotonically in time.

iii.) $A < 0$ and $B > 0$

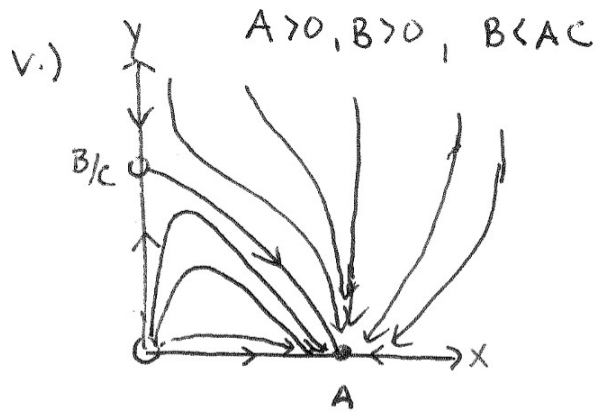


Similar to case ii, only the fixed point $(0, B/c)$ is attractive and only n_2 mode survives asymptotically in time. The number of n_1 photons decreases monotonically in time. For this scenario $G_1 < k_1/N_0$ & $G_2 > k_2/N_0$

iv.) $A > 0$, $B > 0$, $B > AC$



The only attractor is the $(0, B/c)$ fixed point, so again the laser settles into n_2 mode operation. However, the initial transients can now involve a temporary increase in the strength of the n_1 mode. Here $G_1 > k_1/N_0$, $G_2 > k_2/N_0$ and $\frac{G_2 - k_2/N_0}{G_1} > G_2 \frac{G_1 - k_1/N_0}{G_1^2}$



Again there is only one attractor; except now it is $(A, 0)$. $(B, B/c)$ is a saddle, which brings the possibility of transient two mode emission.

This happens for $G_1 - k_1/N_0 > 0$,

$G_2 - k_2/N_0 > 0$ and

$$\frac{G_2 - k_2/N_0}{G_1} < G_2 \frac{G_1 - k_1/N_0}{G_1^2}.$$